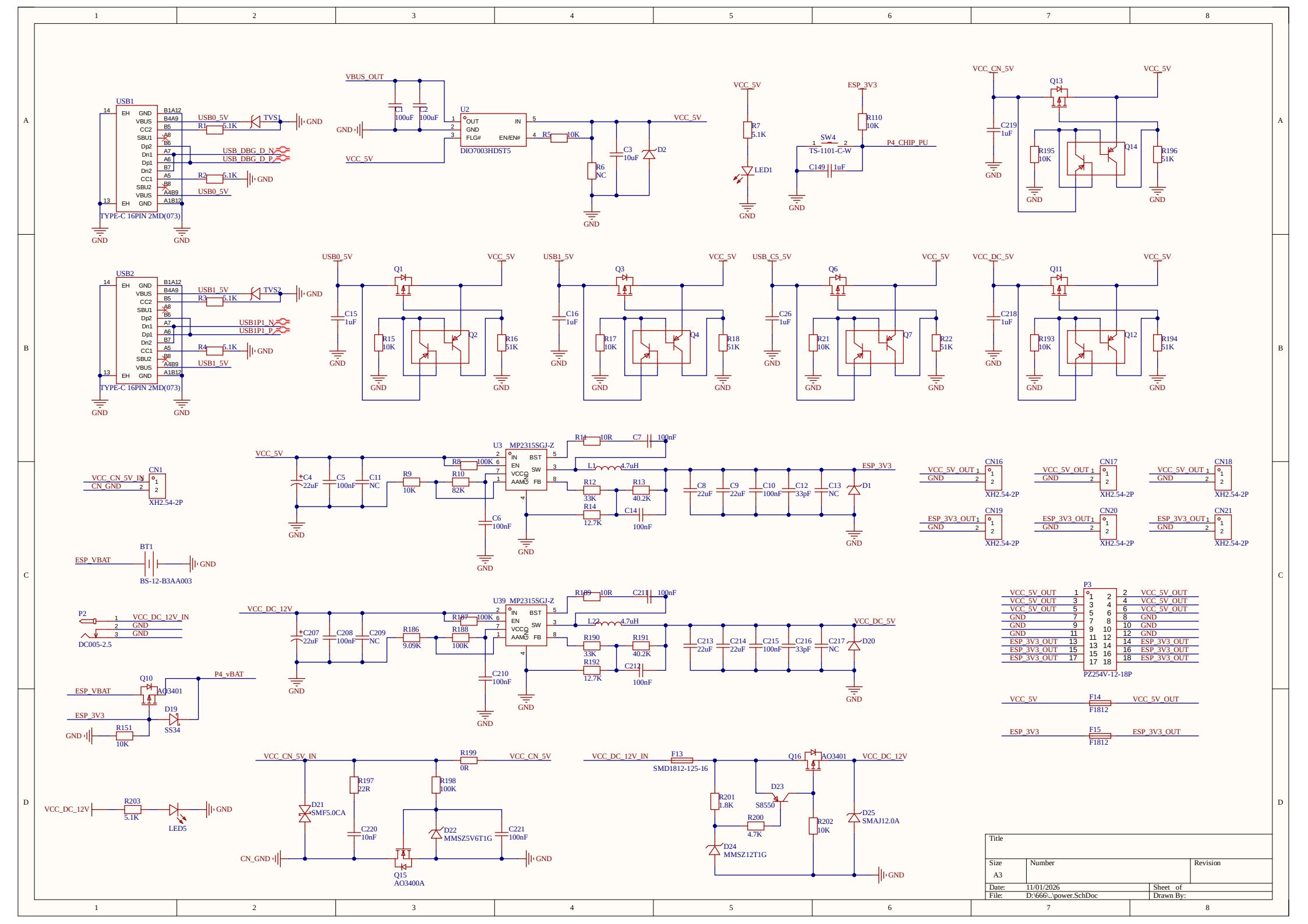
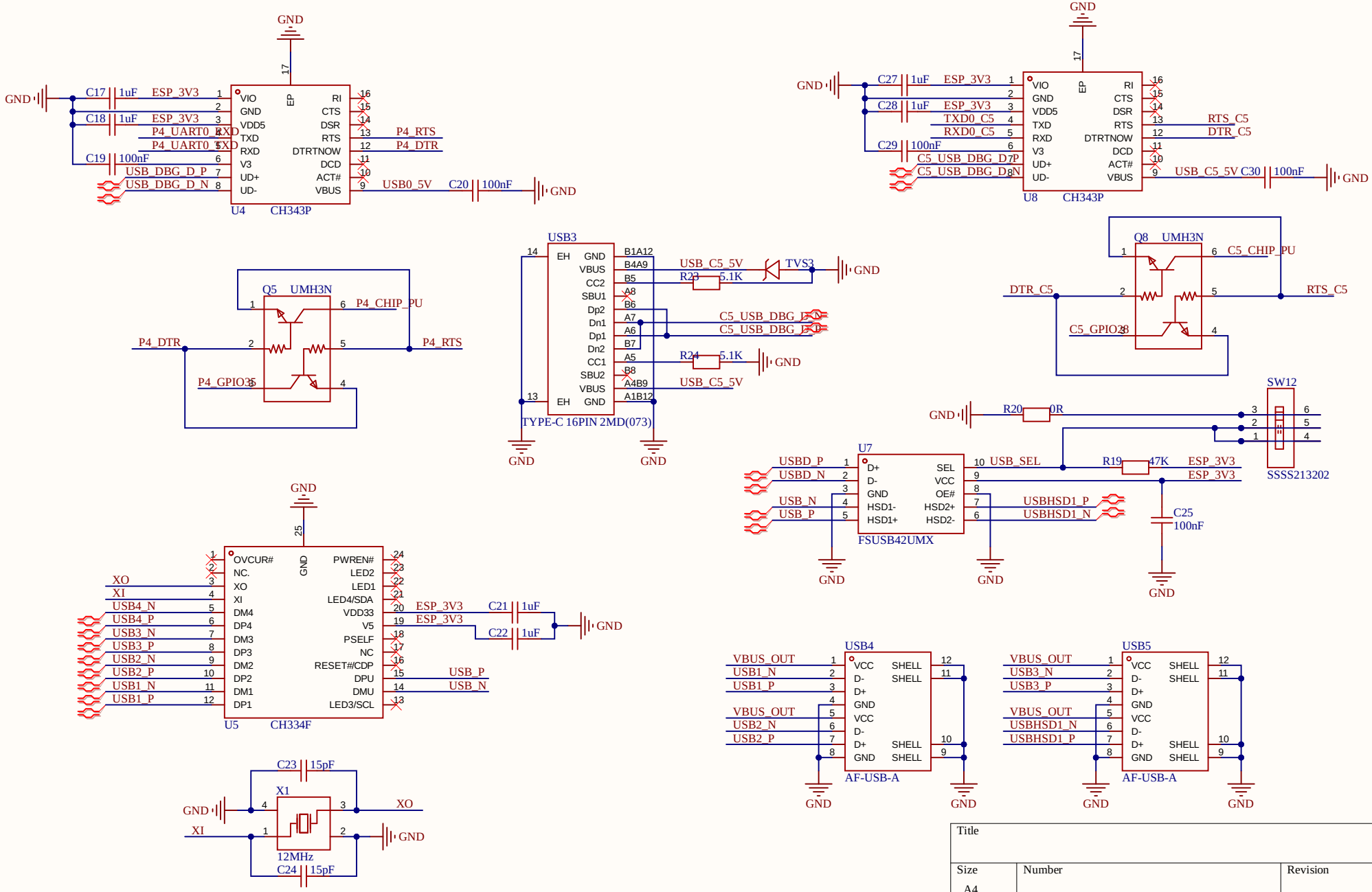
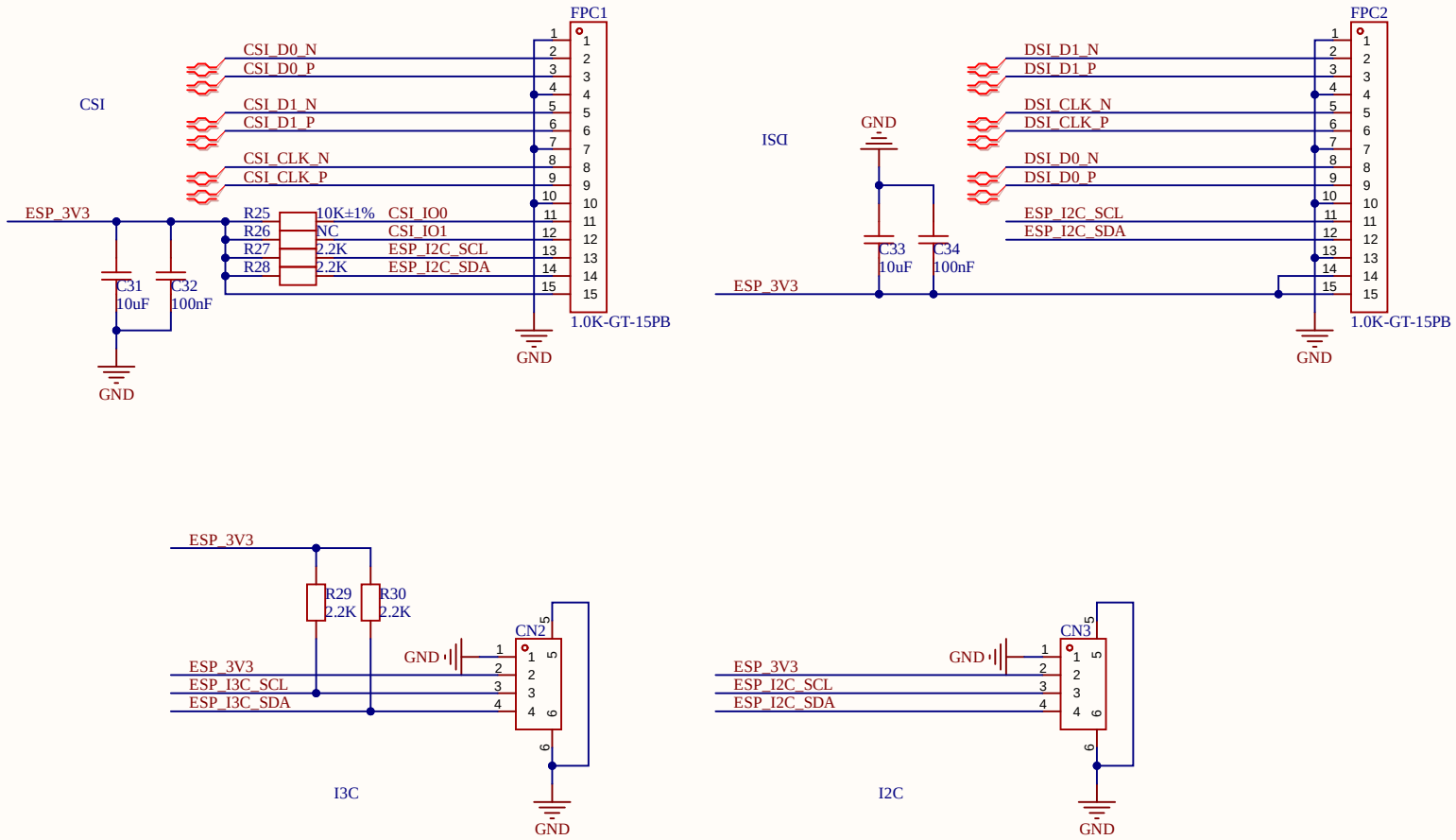
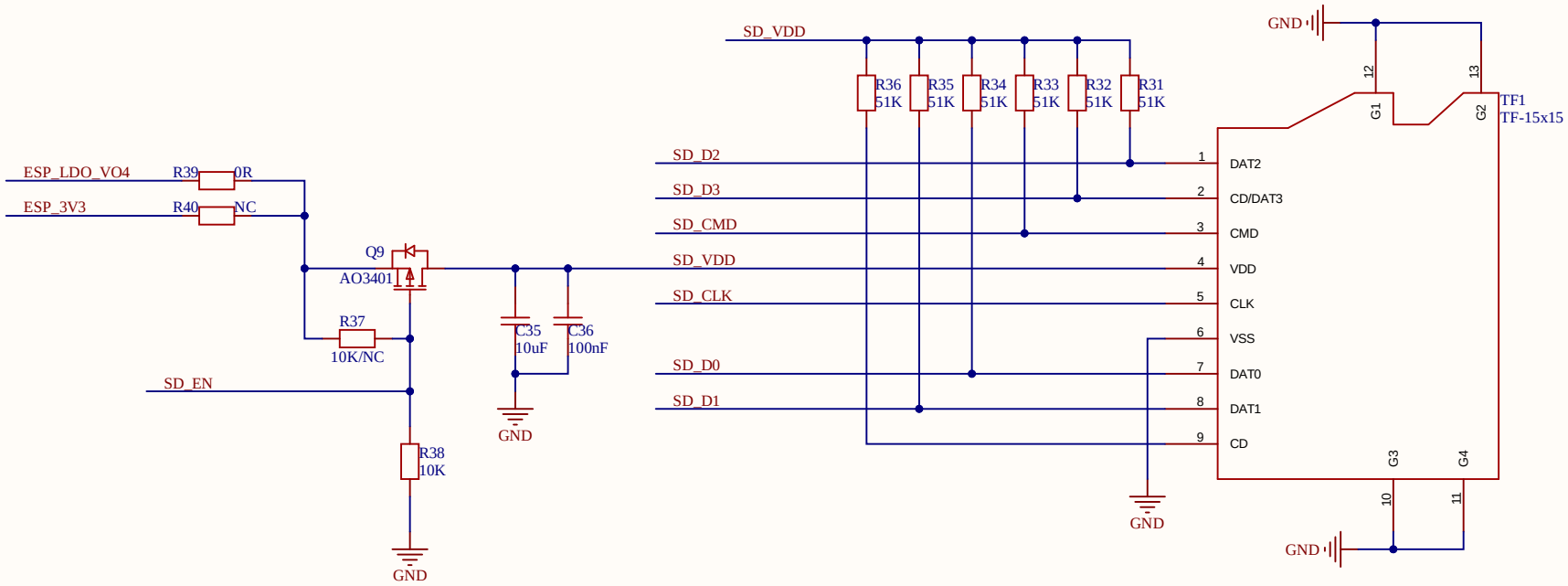

USB1
TYPE-C 16PIN 2MD(073)
USB0_5V
TVS1
USB_DBG_D_N
USB_DBG_D_P
VBUS_OUT
U2
DIO7003HDST5
VCC_5V
ESP_3V3
R7 5.1K
LED1
SW4
TS-1101-C-W
P4_CHIP_PU
R110 10K
C149 1uF
VCC_CN_5V
Q13
Q14
R195 10K
R196 51K
C219 1uF
USB2
USB1_5V
TVS2
USB1P1_N
USB1P1_P
USB0_5V
USB1_5V
USB_C5_5V
VCC_DC_5V
Q1
Q2
Q3
Q4
Q6
Q7
Q11
Q12
R15 10K
R16 51K
R17 10K
R18 51K
R21 10K
R22 51K
R193 10K
R194 51K
C15 1uF
C16 1uF
C26 1uF
C218 1uF
CN1
XH2.54-2P
U3 MP2315SGJ-Z
ESP_3V3
D1
BT1
BS-12-B3AA003
ESP_VBAT
CN16
CN17
CN18
CN19
CN20
CN21
P2
DC005-2.5
VCC_DC_12V_IN
U39 MP2315SGJ-Z
VCC_DC_12V
VCC_DC_5V
D20
Q10
AO3401
P4_vBAT
D19
SS34
R151 10K
P3
PZ254V-12-18P
F14
F15
F1812
R203 5.1K
LED5
VCC_CN_5V_IN
R199 0R
R197 22R
R198 100K
D21 SMF5.0CA
D22 MMSZ5V6T1G
C220 10nF
C221 100nF
CN_GND
Q15 AO3400A
VCC_DC_12V_IN
F13 SMD1812-125-16
Q16 AO3401
R201 1.8K
R200 4.7K
R202 10K
D23 S8550
D24 MMSZ12T1G
D25 SMAJ12.0A
Title
Size A3
Number
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GND
C17 1uF ESP_3V3
C18 1uF ESP_3V3
P4_UART0_RXD
P4_UART0_TXD
C19 100nF
USB_DBG_D_P
USB_DBG_D_N
VIO
GND
VDD5
TXD
RXD
V3
UD+
UD-
RI
CTS
DSR
RTS
DTRTNOW
DCD
ACT#
VBUS
P4_RTS
P4_DTR
USB0_5V
C20 100nF
U4 CH343P
C27 1uF ESP_3V3
C28 1uF ESP_3V3
TXD0_C5
RXD0_C5
C29 100nF
C5_USB_DBG_D_P
C5_USB_DBG_D_N
RTS_C5
DTR_C5
USB_C5_5V
C30 100nF
U8 CH343P
Q5 UMH3N
P4_CHIP_PU
P4_DTR
P4_RTS
P4_GPIO35
USB3
EH
CC2
SBU1
Dp2
Dn1
Dp1
Dn2
CC1
SBU2
B1A12
B4A9
B5
A8
B6
A7
A6
B7
A5
B8
A4B9
A1B12
R23 5.1K
TVS3
C5_USB_DBG_D_N
C5_USB_DBG_D_P
R24 5.1K
TYPE-C 16PIN 2MD(073)
Q8 UMH3N
C5_CHIP_PU
C5_GPIO28
SW12
SSSS213202
R20 0R
U7
USBD_P
USBD_N
USB_N
USB_P
D+
D-
HSD1-
HSD1+
SEL
VCC
OE#
HSD2+
HSD2-
USB_SEL
R19 47K
ESP_3V3
USBHSD1_P
USBHSD1_N
FSUSB42UMX
C25 100nF
OVCUR#
NC.
PWREN#
LED2
LED1
LED4/SDA
VDD33
V5
PSELF
RESET#/CDP
DPU
DMU
LED3/SCL
DM4
DP4
DM3
DP3
DM2
DP2
DM1
DP1
XO
XI
USB4_N
USB4_P
USB3_N
USB3_P
USB2_N
USB2_P
USB1_N
USB1_P
C21 1uF
C22 1uF
U5 CH334F
C23 15pF
X1
12MHz
C24 15pF
USB4
USB5
VBUS_OUT
SHELL
AF-USB-A
Title
Size
A4
Number
Revision
Date: 11/01/2026
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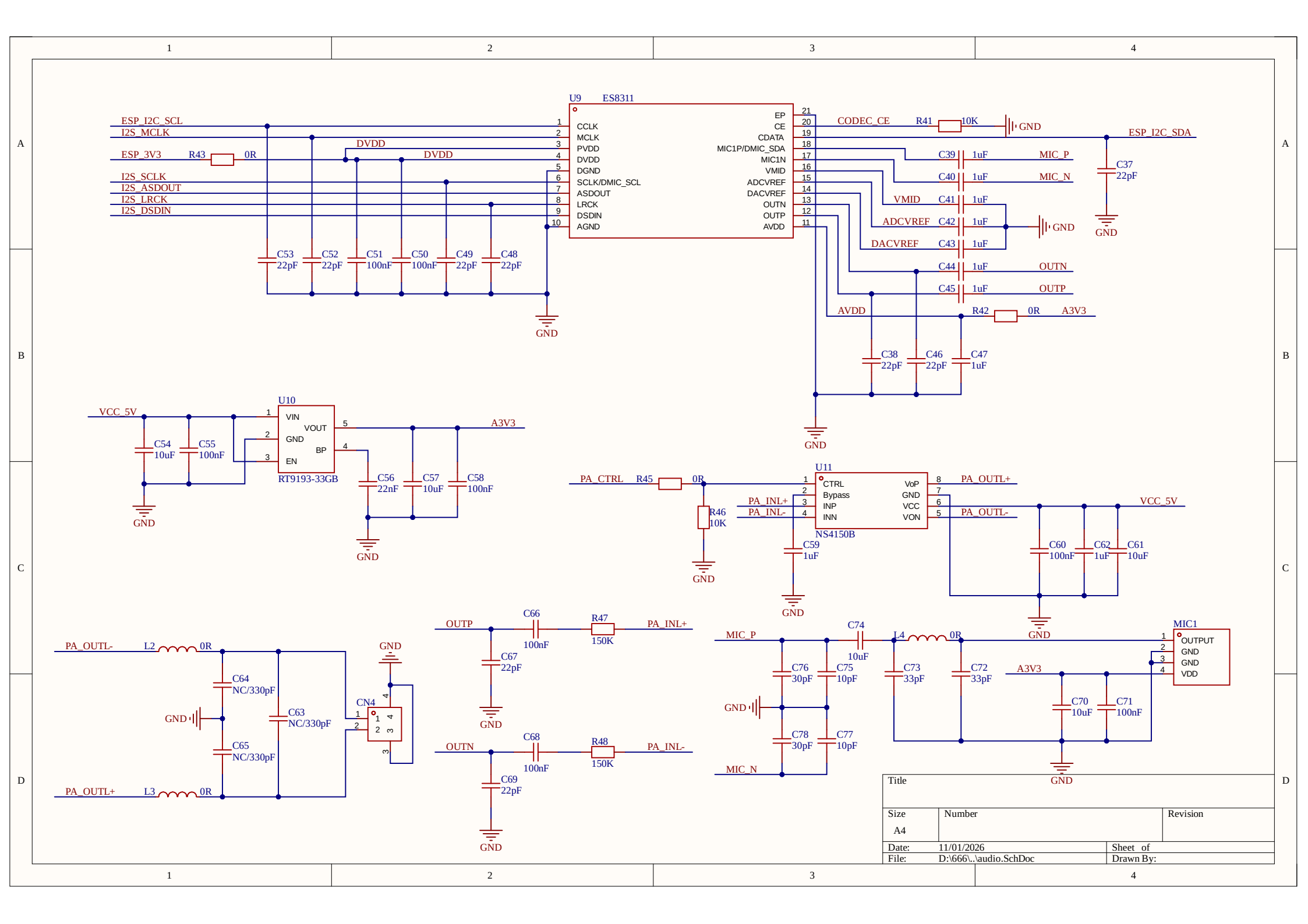


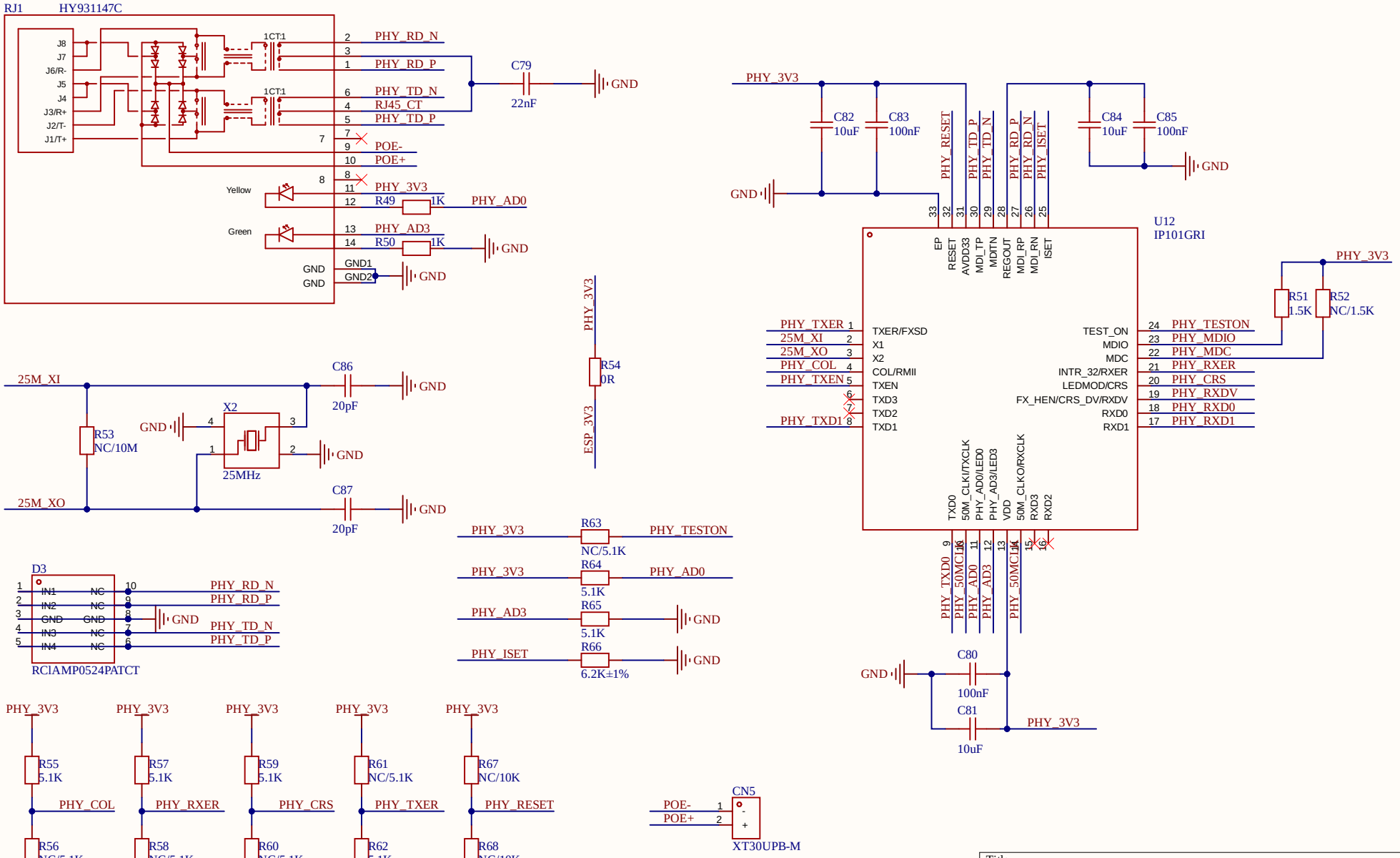
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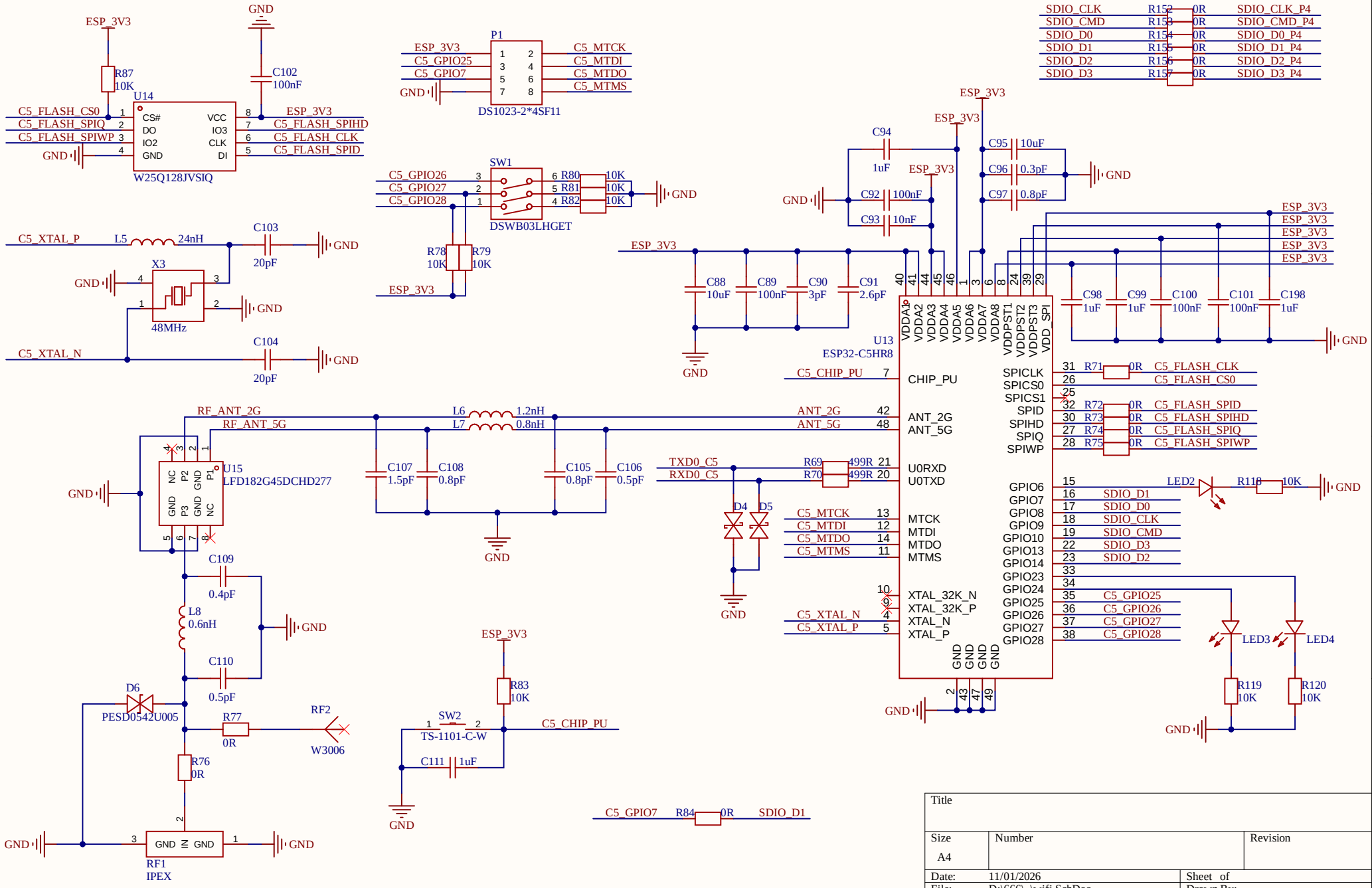


SD_CLK	R158 0R	SD_CLK_P4
SD_CMD	R159 0R	SD_CMD_P4
SD_D0	R160 0R	SD_D0_P4
SD_D1	R161 0R	SD_D1_P4
SD_D2	R162 0R	SD_D2_P4
SD_D3	R163 0R	SD_D3_P4

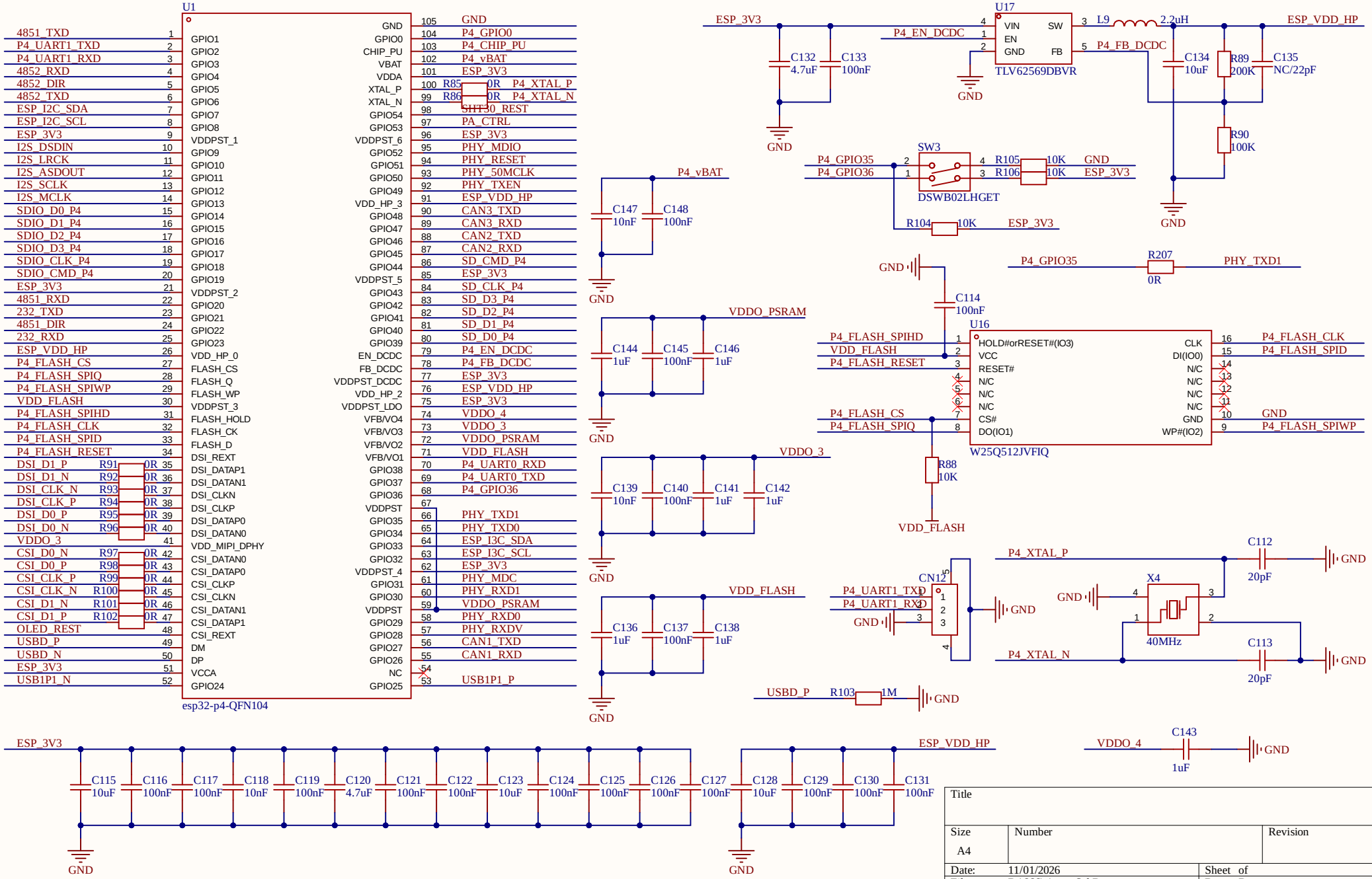
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Date: 11/01/2026		Sheet of
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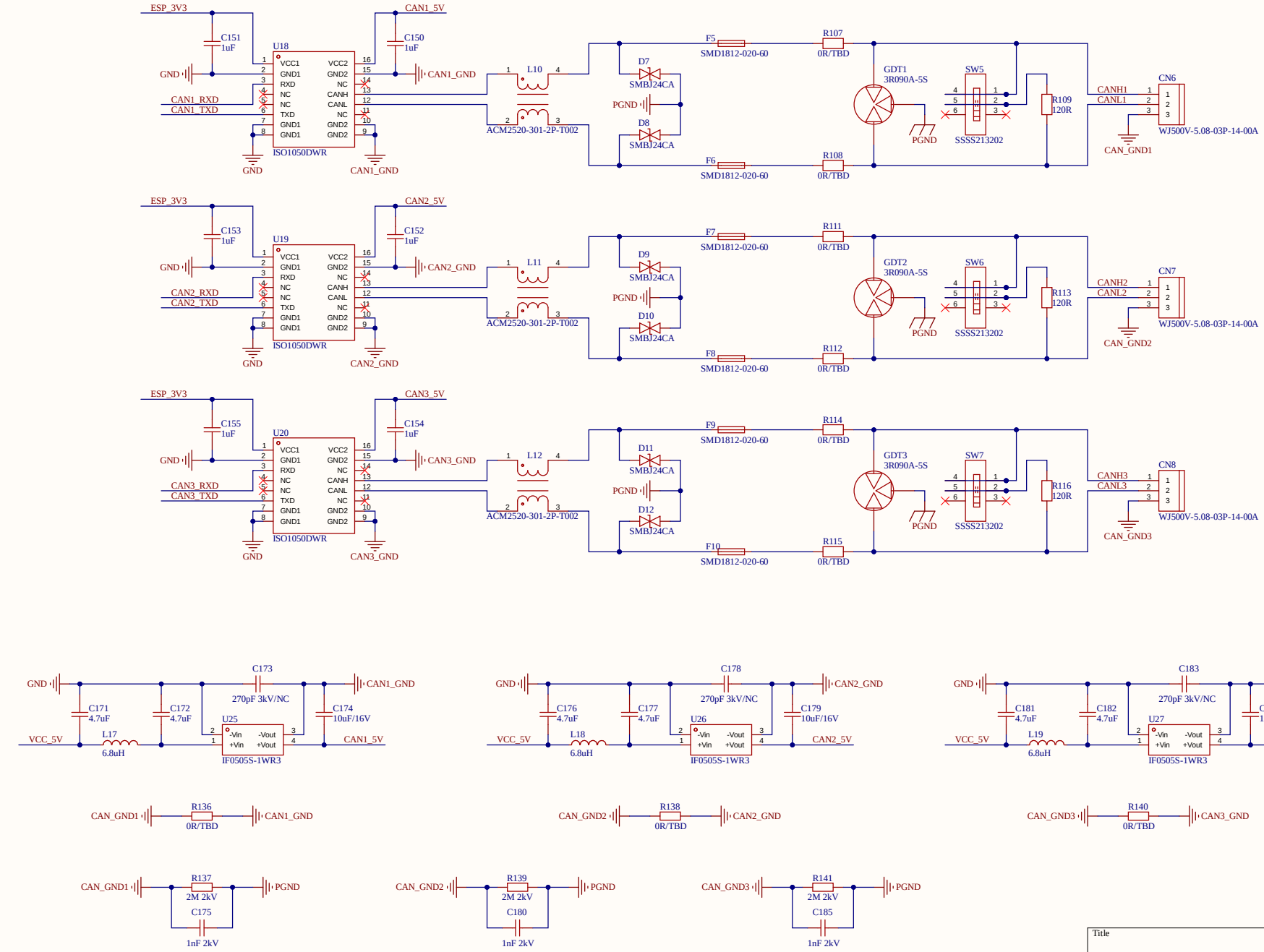

U9 ES8311
ESP_I2C_SCL
I2S_MCLK
ESP_3V3
R43 0R
I2S_SCLK
I2S_ASDOUT
I2S_LRCK
I2S_DSDIN
DVDD
CCLK
MCLK
PVDD
DVDD
DGND
SCLK/DMIC_SCL
ASDOUT
LRCK
DSDIN
AGND
EP
CE
CDATA
MIC1P/DMIC_SDA
MIC1N
VMID
ADCVREF
DACVREF
OUTN
OUTP
AVDD
CODEC_CE
R41 10K
ESP_I2C_SDA
MIC_P
MIC_N
OUTN
OUTP
A3V3
R42 0R
C53 22pF
C52 22pF
C51 100nF
C50 100nF
C49 22pF
C48 22pF
C39 1uF
C40 1uF
C41 1uF
C42 1uF
C43 1uF
C44 1uF
C45 1uF
C37 22pF
C38 22pF
C46 22pF
C47 1uF
GND
U10
VCC_5V
VIN
VOUT
GND
BP
EN
RT9193-33GB
C54 10uF
C55 100nF
C56 22nF
C57 10uF
C58 100nF
A3V3
U11
PA_CTRL
R45 0R
R46 10K
PA_INL+
PA_INL-
CTRL
Bypass
INP
INN
VoP
GND
VCC
VON
NS4150B
PA_OUTL+
PA_OUTL-
C59 1uF
C60 100nF
C62 1uF
C61 10uF
VCC_5V
PA_OUTL-
L2 0R
C64 NC/330pF
C63 NC/330pF
C65 NC/330pF
PA_OUTL+
L3 0R
CN4
OUTP
C66 100nF
R47 150K
PA_INL+
C67 22pF
OUTN
C68 100nF
R48 150K
PA_INL-
C69 22pF
MIC_P
C74 10uF
L4 0R
C76 30pF
C75 10pF
C78 30pF
C77 10pF
MIC_N
C73 33pF
C72 33pF
MIC1
OUTPUT
GND
GND
VDD
A3V3
C70 10uF
C71 100nF
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RJ1 HY931147C
J8
J7
J6/R-
J5
J4
J3/R+
J2/T-
J1/T+
1CT:1
1CT:1
PHY_RD_N
PHY_RD_P
PHY_TD_N
RJ45_CT
PHY_TD_P
POE-
POE+
PHY_3V3
Yellow
Green
R49 1K
R50 1K
PHY_AD0
PHY_AD3
GND
GND1
GND2
C79
22nF
PHY_3V3
C82 10uF
C83 100nF
C84 10uF
C85 100nF
PHY_RESET
PHY_TD_P
PHY_TD_N
PHY_RD_P
PHY_RD_N
PHY_ISET
U12
IP101GRI
EP
RESET
AVDD33
MDI_TP
MDIN
REGOUT
MDI_RP
MDI_RN
ISET
PHY_TXER
25M_XI
25M_XO
PHY_COL
PHY_TXEN
PHY_TXD1
TXER/FXSD
X1
X2
COL/RMII
TXEN
TXD3
TXD2
TXD1
TEST_ON
MDIO
MDC
INTR_32/RXER
LEDMOD/CRS
FX_HEN/CRS_DV/RXDV
RXD0
RXD1
PHY_TESTON
PHY_MDIO
PHY_MDC
PHY_RXER
PHY_CRS
PHY_RXDV
PHY_RXD0
PHY_RXD1
R51 1.5K
R52 NC/1.5K
TXD0
50M_CLKI/TXCLK
PHY_AD0/LED0
PHY_AD3/LED3
VDD
50M_CLKO/RXCLK
RXD3
RXD2
PHY_TXD0
PHY_50MCLKI
PHY_AD0
PHY_AD3
PHY_50MCLKO
C80 100nF
C81 10uF
25M_XI
25M_XO
C86 20pF
C87 20pF
X2
25MHz
R53 NC/10M
R54 0R
ESP_3V3
D3
RClAMP0524PATCT
IN1
IN2
GND
IN3
IN4
NC
R63 NC/5.1K
R64 5.1K
R65 5.1K
R66 6.2K±1%
PHY_ISET
R55 5.1K
R57 5.1K
R59 5.1K
R61 NC/5.1K
R67 NC/10K
R56 NC/5.1K
R58 NC/5.1K
R60 NC/5.1K
R62 5.1K
R68 NC/10K
PHY_COL
PHY_RXER
PHY_CRS
PHY_TXER
PHY_RESET
CN5
XT30UPB-M
Title
Size A4
Number
Revision
Date: 11/01/2026
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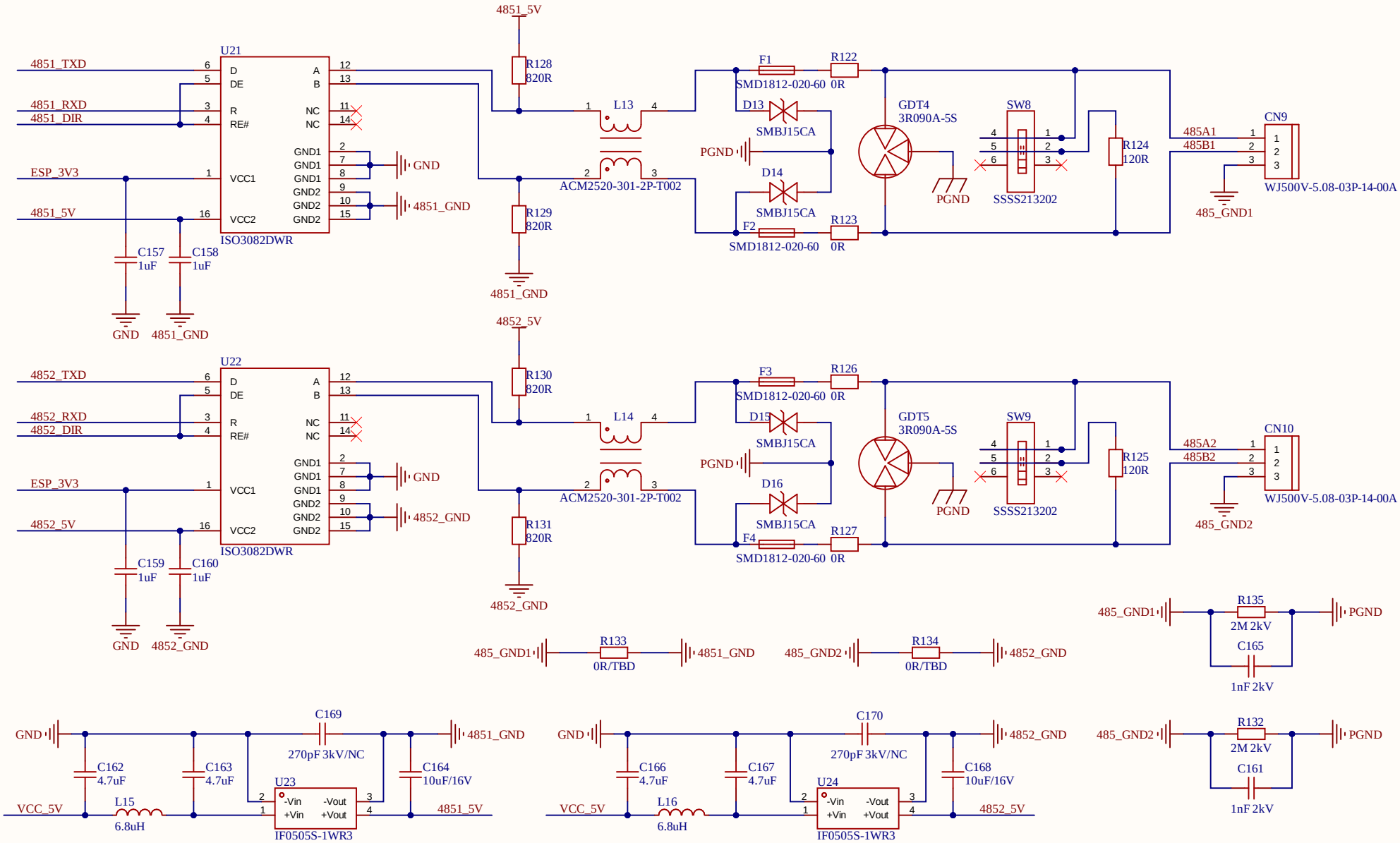


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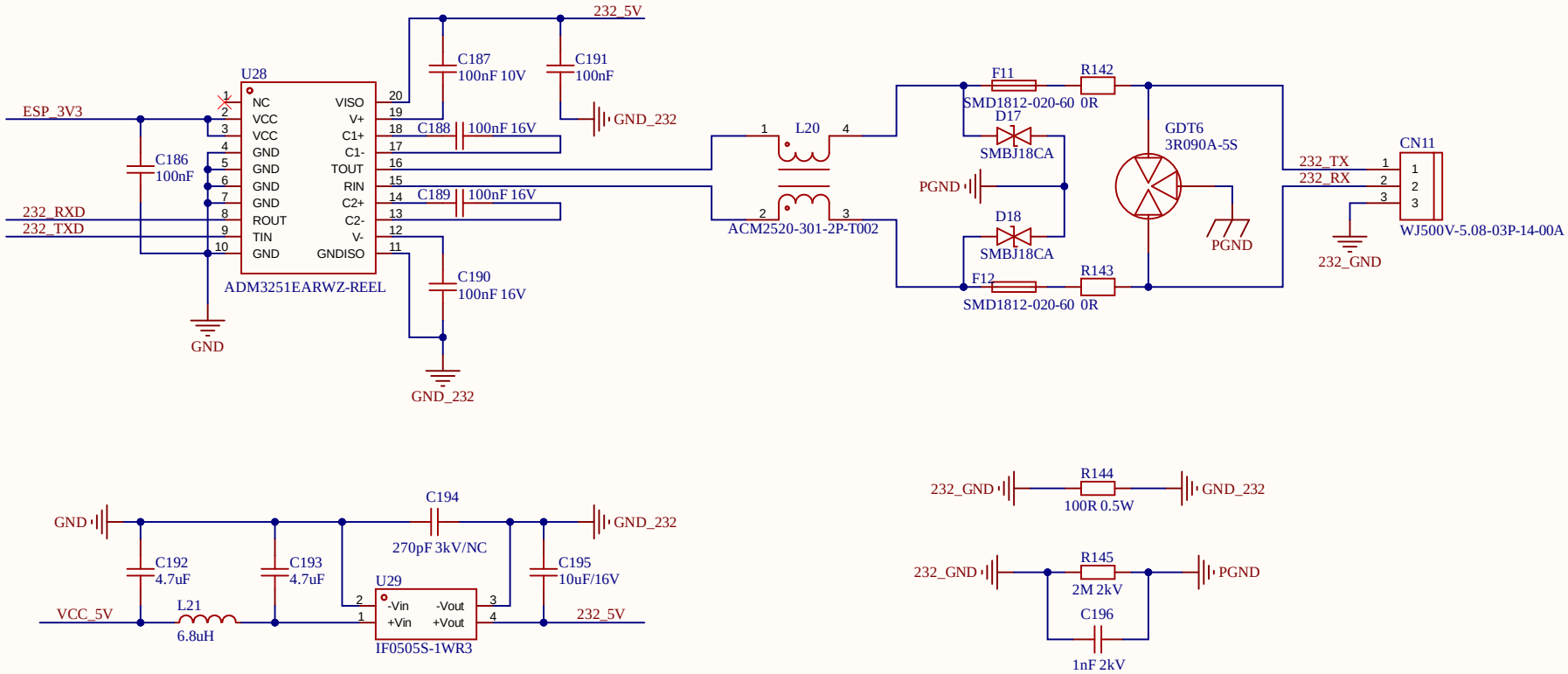

U1
esp32-p4-QFN104
4851_TXD
P4_UART1_TXD
P4_UART1_RXD
4852_RXD
4852_DIR
4852_TXD
ESP_I2C_SDA
ESP_I2C_SCL
ESP_3V3
I2S_DSDIN
I2S_LRCK
I2S_ASDOUT
I2S_SCLK
I2S_MCLK
SDIO_D0_P4
SDIO_D1_P4
SDIO_D2_P4
SDIO_D3_P4
SDIO_CLK_P4
SDIO_CMD_P4
ESP_3V3
4851_RXD
232_TXD
4851_DIR
232_RXD
ESP_VDD_HP
P4_FLASH_CS
P4_FLASH_SPIQ
P4_FLASH_SPIWP
VDD_FLASH
P4_FLASH_SPIHD
P4_FLASH_CLK
P4_FLASH_SPID
P4_FLASH_RESET
DSI_D1_P R91 0R
DSI_D1_N R92 0R
DSI_CLK_N R93 0R
DSI_CLK_P R94 0R
DSI_D0_P R95 0R
DSI_D0_N R96 0R
VDDO_3
CSI_D0_N R97 0R
CSI_D0_P R98 0R
CSI_CLK_P R99 0R
CSI_CLK_N R100 0R
CSI_D1_N R101 0R
CSI_D1_P R102 0R
OLED_REST
USBD_P
USBD_N
ESP_3V3
USB1P1_N
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
VDDPST_1
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
VDDPST_2
GPIO20
GPIO21
GPIO22
GPIO23
VDD_HP_0
FLASH_CS
FLASH_Q
FLASH_WP
VDDPST_3
FLASH_HOLD
FLASH_CK
FLASH_D
DSI_REXT
DSI_DATAP1
DSI_DATAN1
DSI_CLKN
DSI_CLKP
DSI_DATAP0
DSI_DATAN0
VDD_MIPI_DPHY
CSI_DATAN0
CSI_DATAP0
CSI_CLKP
CSI_CLKN
CSI_DATAN1
CSI_DATAP1
CSI_REXT
DM
DP
VCCA
GPIO24
GND 105
GPIO0 104
CHIP_PU 103
VBAT 102
VDDA 101
XTAL_P 100
XTAL_N 99
GPIO54 98
GPIO53 97
VDDPST_6 96
GPIO52 95
GPIO51 94
GPIO50 93
GPIO49 92
VDD_HP_3 91
GPIO48 90
GPIO47 89
GPIO46 88
GPIO45 87
GPIO44 86
VDDPST_5 85
GPIO43 84
GPIO42 83
GPIO41 82
GPIO40 81
GPIO39 80
EN_DCDC 79
FB_DCDC 78
VDDPST_DCDC 77
VDD_HP_2 76
VDDPST_LDO 75
VFB/VO4 74
VFB/VO3 73
VFB/VO2 72
VFB/VO1 71
GPIO38 70
GPIO37 69
GPIO36 68
VDDPST 67
GPIO35 66
GPIO34 65
GPIO33 64
GPIO32 63
VDDPST_4 62
GPIO31 61
GPIO30 60
VDDPST 59
GPIO29 58
GPIO28 57
GPIO27 56
GPIO26 55
NC 54
GPIO25 53
GND
P4_GPIO0
P4_CHIP_PU
P4_vBAT
ESP_3V3
R85 0R P4_XTAL_P
R86 0R P4_XTAL_N
SHT30_REST
PA_CTRL
ESP_3V3
PHY_MDIO
PHY_RESET
PHY_50MCLK
PHY_TXEN
ESP_VDD_HP
CAN3_TXD
CAN3_RXD
CAN2_TXD
CAN2_RXD
SD_CMD_P4
ESP_3V3
SD_CLK_P4
SD_D3_P4
SD_D2_P4
SD_D1_P4
SD_D0_P4
P4_EN_DCDC
P4_FB_DCDC
ESP_3V3
ESP_VDD_HP
ESP_3V3
VDDO_4
VDDO_3
VDDO_PSRAM
VDD_FLASH
P4_UART0_RXD
P4_UART0_TXD
P4_GPIO36
PHY_TXD1
PHY_TXD0
ESP_I3C_SDA
ESP_I3C_SCL
ESP_3V3
PHY_MDC
PHY_RXD1
VDDO_PSRAM
PHY_RXD0
PHY_RXDV
CAN1_TXD
CAN1_RXD
USB1P1_P
C147 10nF
C148 100nF
C144 1uF
C145 100nF
C146 1uF
C139 10nF
C140 100nF
C141 1uF
C142 1uF
C136 1uF
C137 100nF
C138 1uF
U17
TLV62569DBVR
VIN
SW
EN
GND
FB
L9 2.2uH
P4_FB_DCDC
C132 4.7uF
C133 100nF
C134 10uF
R89 200K
C135 NC/22pF
R90 100K
SW3
DSWB02LHGET
P4_GPIO35
P4_GPIO36
R105 10K GND
R106 10K ESP_3V3
R104 10K ESP_3V3
R207 0R
C114 100nF
U16
W25Q512JVFIQ
HOLD#orRESET#(IO3)
VCC
RESET#
N/C
CS#
DO(IO1)
CLK
DI(IO0)
WP#(IO2)
P4_FLASH_CLK
P4_FLASH_SPID
P4_FLASH_SPIWP
R88 10K
CN12
P4_UART1_TXD
P4_UART1_RXD
P4_XTAL_P
P4_XTAL_N
X4
40MHz
C112 20pF
C113 20pF
USBD_P R103 1M
C115 10uF
C116 100nF
C117 100nF
C118 10nF
C119 100nF
C120 4.7uF
C121 100nF
C122 100nF
C123 10uF
C124 100nF
C125 100nF
C126 100nF
C127 100nF
C128 10uF
C129 100nF
C130 100nF
C131 100nF
C143 1uF
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Number
Revision
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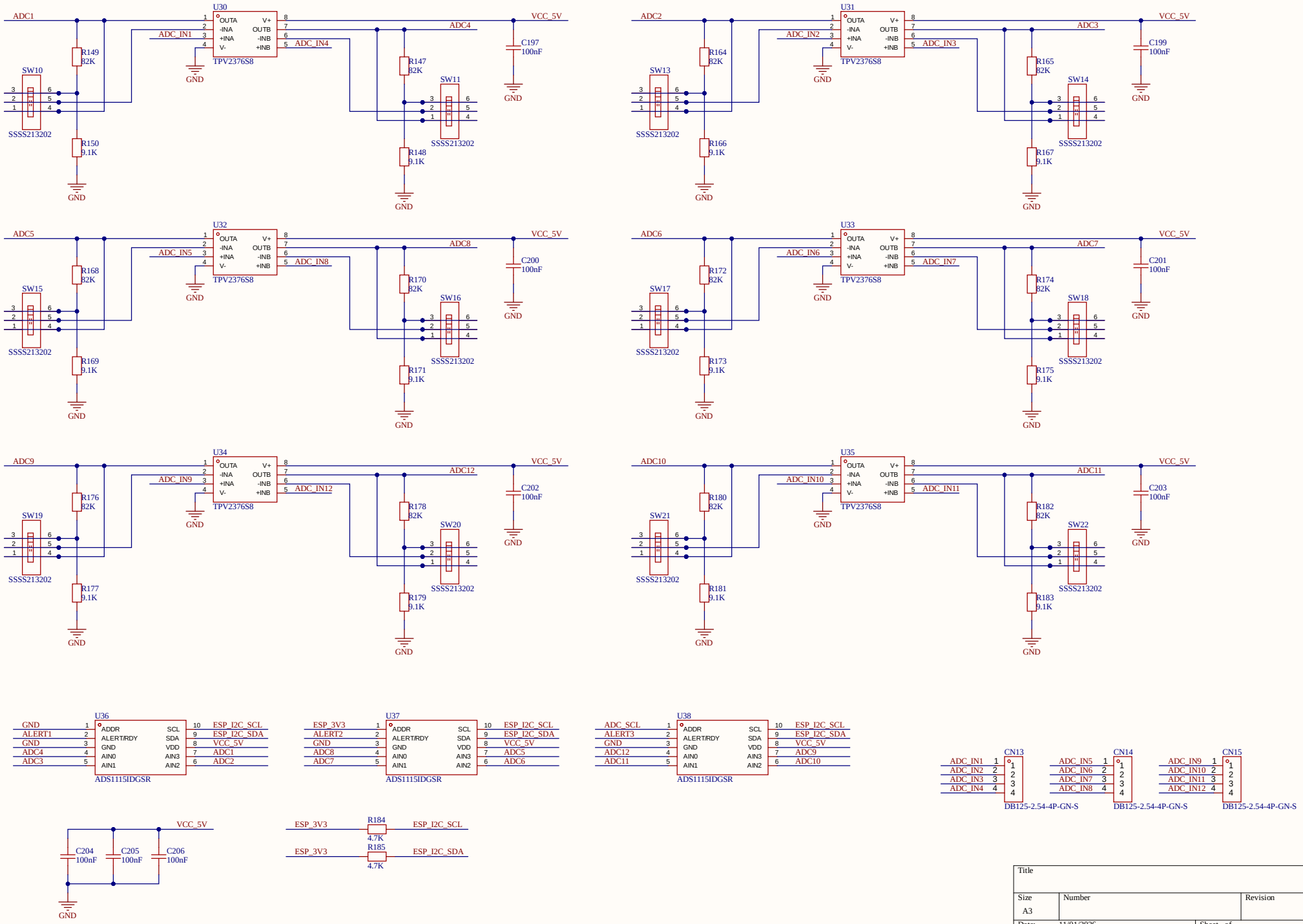
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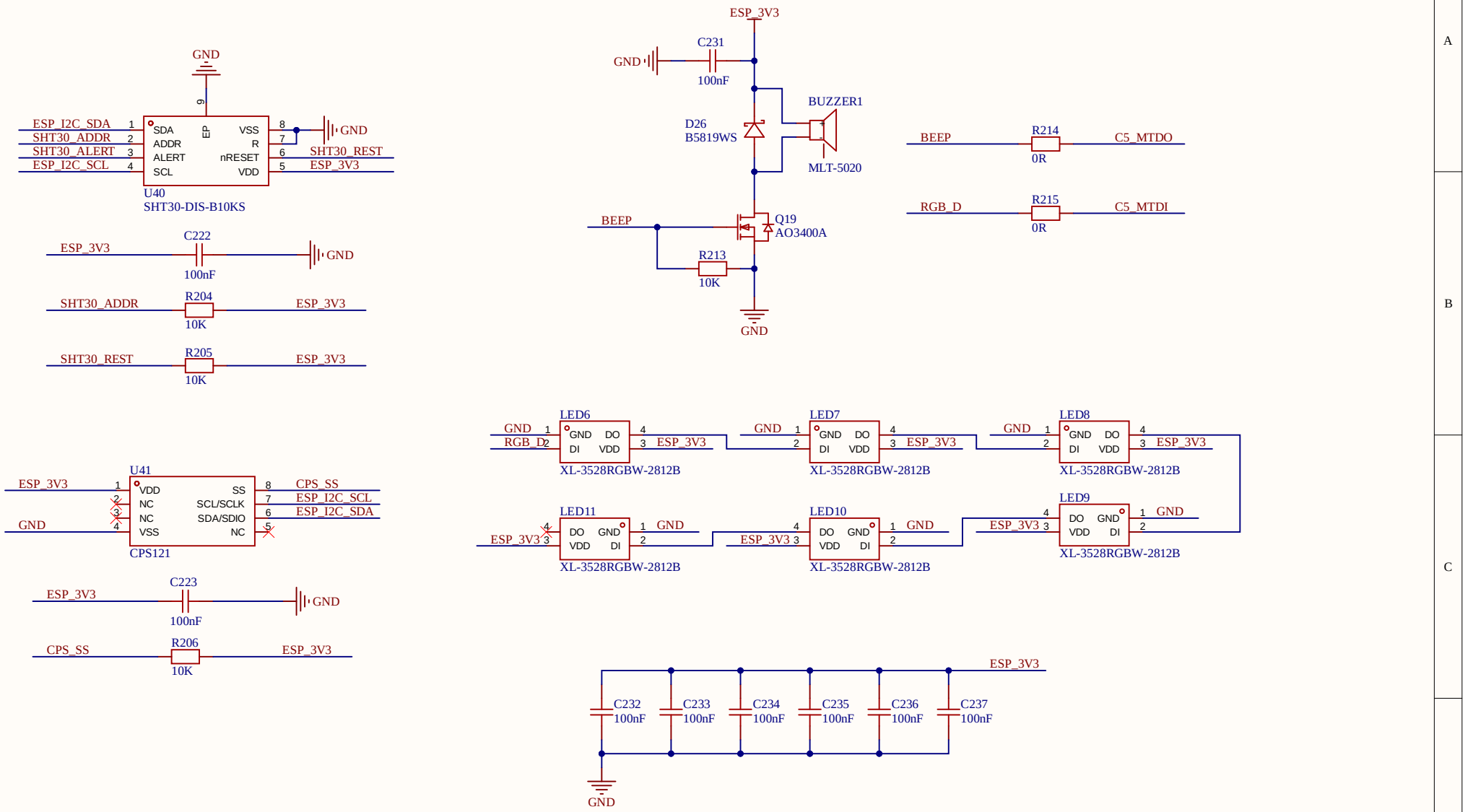


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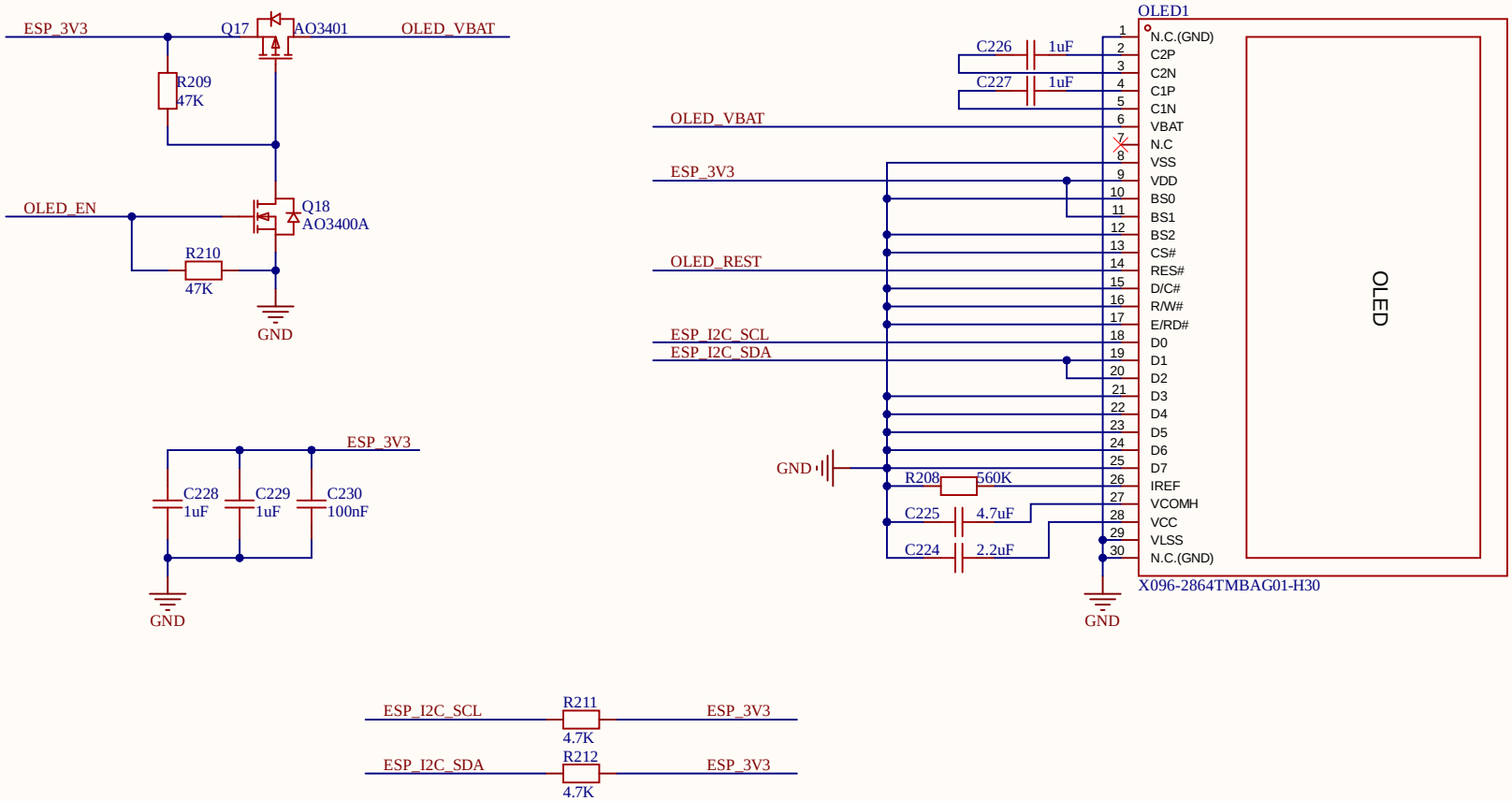


Title		
Size: A4	Number	Revision
Date: 11/01/2026		Sheet of
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ADC1
U30
TPV2376S8
ADC_IN1
ADC_IN4
ADC4
VCC_5V
C197
100nF
R149
82K
SW10
SSSS213202
R150
9.1K
R147
82K
SW11
SSSS213202
R148
9.1K
GND
ADC2
U31
TPV2376S8
ADC_IN2
ADC_IN3
ADC3
C199
100nF
R164
82K
SW13
SSSS213202
R166
9.1K
R165
82K
SW14
SSSS213202
R167
9.1K
ADC5
U32
TPV2376S8
ADC_IN5
ADC_IN8
ADC8
C200
100nF
R168
82K
SW15
SSSS213202
R169
9.1K
R170
82K
SW16
SSSS213202
R171
9.1K
ADC6
U33
TPV2376S8
ADC_IN6
ADC_IN7
ADC7
C201
100nF
R172
82K
SW17
SSSS213202
R173
9.1K
R174
82K
SW18
SSSS213202
R175
9.1K
ADC9
U34
TPV2376S8
ADC_IN9
ADC_IN12
ADC12
C202
100nF
R176
82K
SW19
SSSS213202
R177
9.1K
R178
82K
SW20
SSSS213202
R179
9.1K
ADC10
U35
TPV2376S8
ADC_IN10
ADC_IN11
ADC11
C203
100nF
R180
82K
SW21
SSSS213202
R181
9.1K
R182
82K
SW22
SSSS213202
R183
9.1K
U36
ADS1115IDGSR
GND
ALERT1
GND
ADC4
ADC3
ESP_I2C_SCL
ESP_I2C_SDA
VCC_5V
ADC1
ADC2
U37
ADS1115IDGSR
ESP_3V3
ALERT2
GND
ADC8
ADC7
ESP_I2C_SCL
ESP_I2C_SDA
VCC_5V
ADC5
ADC6
U38
ADS1115IDGSR
ADC_SCL
ALERT3
GND
ADC12
ADC11
ESP_I2C_SCL
ESP_I2C_SDA
VCC_5V
ADC9
ADC10
CN13
DB125-2.54-4P-GN-S
ADC_IN1
ADC_IN2
ADC_IN3
ADC_IN4
CN14
DB125-2.54-4P-GN-S
ADC_IN5
ADC_IN6
ADC_IN7
ADC_IN8
CN15
DB125-2.54-4P-GN-S
ADC_IN9
ADC_IN10
ADC_IN11
ADC_IN12
VCC_5V
C204
100nF
C205
100nF
C206
100nF
ESP_3V3
R184
4.7K
ESP_I2C_SCL
ESP_3V3
R185
4.7K
ESP_I2C_SDA
Title
Size
A3
Number
Revision
Date:
11/01/2026
Sheet of
File:
D:\666\..\adc.SchDoc
Drawn By:



Title		
Size A4	Number	Revision
Date: 11/01/2026		Sheet of
File: D:\666\..\sensor.SchDoc		Drawn By:



Title		
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Date: 11/01/2026		Sheet of
File: D:\666\..\OLED.SchDoc		Drawn By:

4mil	90ohm	7mil
single	50ohm	13.8mil
4mil	100ohm	4.7mil

